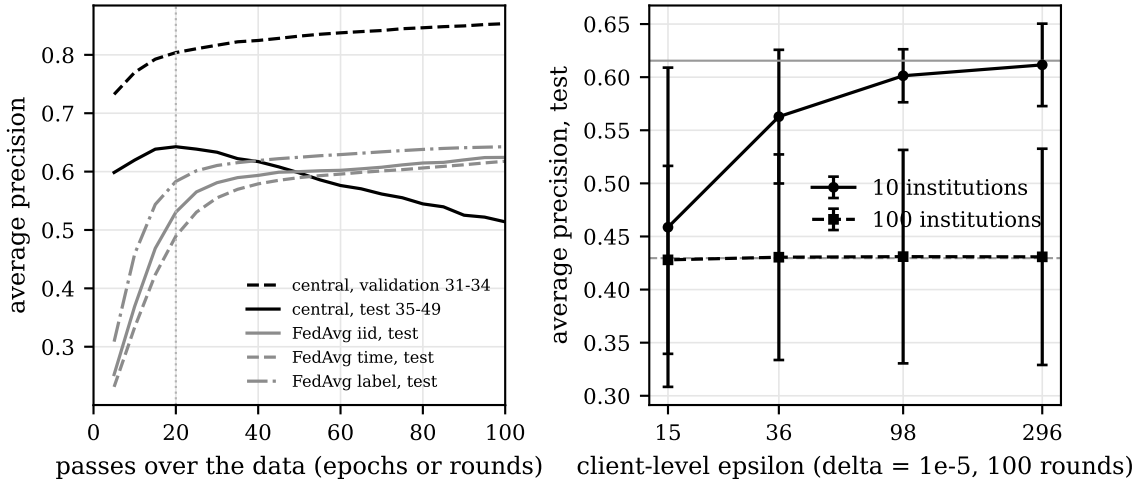
(a) learning curves
average precision
0.8
0.7
0.6
0.5
0.4
0.3
central, validation 31-34
central, test 35-49
FedAvg iid, test
FedAvg time, test
FedAvg label, test
0
20
40
60
80
100
passes over the data (epochs or rounds)
(b) cost of differential privacy
average precision, test
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0.30
10 institutions
100 institutions
15
36
98
296
client-level epsilon (delta = 1e-5, 100 rounds)